

Experiment – 17
CSA1717 - Artificial Intelligence

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Aim:

To write a Python program to illustrate different set operations such as union, intersection, difference, and symmetric difference.

Algorithm:

1. Start the program.
2. Create two sets, A and B.
3. Perform the following operations:
 - **Union:** Combine all elements from both sets.
 - **Intersection:** Find common elements between sets.
 - **Difference:** Find elements present in one set but not in the other.
 - **Symmetric Difference:** Find elements that are in either set but not in both.
4. Display the results of each operation.
5. Stop the program.

Source code:

```
# Program to illustrate different set operations
```

```
# Define sets
```

```
A = {1, 2, 3, 4, 5}
```

```
B = {4, 5, 6, 7, 8}
```

```
# Display sets
```

```
print("Set A:", A)
```

```
print("Set B:", B)
```

```
# Union
```

```
print("\nUnion of A and B:", A | B)
```

```
# Intersection
```

```
print("Intersection of A and B:", A & B)
```

```
# Difference
```

```
print("Difference of A and B:", A - B)
```

```
# Symmetric Difference
```

```
print("Symmetric Difference of A and B:", A ^ B)
```

Flowchart:



Output:

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:1)
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/hemal/OneDrive/Documents/SIMATS/AI/A/
Set A: {1, 2, 3, 4, 5}
Set B: {4, 5, 6, 7}

Union (A | B): {1, 2, 3, 4, 5, 6, 7}
Intersection (A & B): {4, 5}
Difference (A - B): {1, 2, 3}
Difference (B - A): {6, 7}
Symmetric Difference (A ^ B): {1, 2, 3, 6, 7}

Is 3 in A? True
Is 10 in B? False

Is A subset of B? False
Is A superset of B? False

After adding 10 to A: {1, 2, 3, 4, 5, 10}
After removing 2 from A: {1, 3, 4, 5, 10}
After discarding 100 (no error): {1, 3, 4, 5, 10}
```

Result:

The program successfully demonstrates different set operations union, intersection, difference, and symmetric difference using Python's built-in set operators.